Abhishek Sharma

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Experience

- **QuaEra Insights** — New York, NY (Remote)
  Data Scientist — *May 2025 – Present*
  - **Creative Recommendation**: Developed personalized direct-mail and email recommendation models across finance, healthcare, and nonprofit clients, contributing to an average campaign response growth from 1% to 3%
  - **Customer History Modeling**: Built PyTorch models using customer interactions, response history; achieved 0.860 out-of-time ROC-AUC using two-stage training with a frozen base network and delta network.
  - **Funnel and Value Models**: Developed email engagement classifiers and balance-change regression models; benchmarked against LightGBM, XGBoost, CatBoost, and hurdle models.
  - **LLM Feature Engineering**: Built an LLM pipeline that turns ad creatives into a "genetic signature" of theme and offer-term features. Devised a hybrid greedy and random-forest algorithm to select the best features for model training and dataset enrichment.
  - **Model Evaluation**: Implemented isotonic and spline calibration, evaluated ROC-AUC and segment-level lift on out-of-time data, and tracked experiments using MLflow.
  - **ML Pipelines**: Built configuration-driven Databricks/Delta Lake pipelines for feature engineering, model training, parallel experiments, monthly scoring, and personalized campaign assignments.
- **MacroXStudio, Inc.** — San Francisco, CA
  Data Scientist — *Summer 2024*
  - **Data Analytics Chatbot**: Business-facing product that answers natural-language questions about company data by generating and running Python code. Built on Google Cloud APIs with a Streamlit interface; about 3-second responses.
  - **LLM Benchmarking**: Compared four LLM pipelines for different data analysis steps; chose fine-tuned LLaMA for column identification and GPT-4o for code generation, reaching 90% accuracy on produced output generated
- **University of Rochester** — Rochester, NY
  Research Assistant — *Oct 2023 – Jan 2025*
  - **PlasAnn**: Curated plasmid-specific database and annotation pipeline for standardized gene and function analysis. Built Python/R data pipelines on a Linux HPC cluster. Co-author, *Nucleic Acids Research* (2026).
  - **Plasmid Ecology**: Study showing that plasmid prevalence is independent of antibiotic resistance in environmental Enterobacteriaceae. Co-author, *Microbial Genomics* (2025).
  - **Generative DNA Models**: Explored transformer-based diffusion models for generating synthetic DNA sequences.
- **Vedanta Limited** — India
  Business Analyst — *Jul 2021 – Jun 2023*
  - **Procurement Analytics**: Market analysis, time-series forecasting, and Power BI dashboards that guided supplier selection. Contributed to over $18M in FY2022–23 savings; received CEO Appreciation Award.
  - **Energy Forecasting**: Neural-network and SARIMAX electricity-price forecasts that informed plant operations, supporting 5% higher boiler efficiency and 4.5% lower raw-material use.

Technical Skills

- **Languages**: Python, R, C++, SQL
  ML / NLP: PyTorch, scikit-learn, LightGBM, XGBoost, CatBoost, Transformers, LLMs (GPT-4o, LLaMA fine-tuning), MLflow
  Data and Cloud: Databricks (Jobs, Delta Lake), Spark, AWS S3, Azure Blob Storage, Google Cloud, Linux/HPC, Git, Streamlit, Power BI
  Methods: Deep Learning, Sequential Modeling, Feature Selection, Model Calibration, Time-Series Forecasting, Feature Engineering, ETL, Statistical Testing

Education

- **University of Rochester** — Rochester, NY
  M.S. in Data Science; GPA: 3.98/4.00 — *Aug 2023 – Dec 2024*
- **Indian Institute of Technology Kanpur** — Kanpur, India
  B.Tech. in Chemical Engineering; Merit Tuition Scholarship — *Aug 2017 – Jul 2021*